Owain **West**

owest@sas.upenn.edu | owainwest.com | owainwest | owainwest

Owain is ideal for high-velocity roles, bringing particular expertise in data engineering, ethical AI, and value-driven project delivery.

Experience

TickPick | Senior Data Engineer New York | July 2026 - Present

- Responsible for engineering delivery of TickPick's core internal AI-enablement platform spanning Data, Engineering, and Product teams.

Beyond Identity | Data Engineer II New York | May 2024-June 2026

- Built and drove adoption of typed spec deliverables system for agentic developer workflows, enabling process auditability of AI output.
- Led implementation of secure credential MCP credential escrow for the Ceros agentic security harness.
- Built high-performance Rust+Kafka-based microservices (Tokio, Axum, rdkafka) to support SIEM integration with Elasticsearch, Splunk, Datadog, Sentinel, and other observability platforms. Shipped with 95% reduction in p90 processing latency compared to legacy implementation.
- Developed real-time threat-signal models and Policy Integration Points (PIPs) in Go and Rust within a zero-trust authentication framework, including logic for IP novelty and travel-speed heuristics.
- Delivered Rust-based, real-time, thread-safe log and metric aggregation infrastructure, powering the new "SIEM Logs" platform feature.
- Championed standardized OpenAPI-driven service generation workflows using Rust (Axum), cutting new microservice ramp-up time by 85%.
- Managed AWS infrastructure for Rust/Go services with Terraform, k8s, and Gitlab CI/CD, using Snowflake, Postgres, and dbt for data workloads.

TickPick | Data Engineer New York | May 2022 - May 2024

- Led TickPick's Data Engineering initiatives using Snowflake, Databricks, and Azure ML; orchestrating data flows across hundreds of tables.
- Pioneered inbound automated ticket transfers, fundamentally enabling "autotransfer" app feature launch.
- Drove data engineering implementation of margin optimization pipe, materially boosting relative conversion rate.
- Automated pricing comparisons against 100k+ competitor listings daily, reinforcing company's "price dominance" strategy.
- Mentored junior engineer while managing 400+ data pipelines loading hundreds of millions of rows monthly.
- Negotiated with ETL vendors, in one case reducing annual costs by over 70% through strategic selection and cost-benefit analysis.
- Created Streamlit user interfaces for stakeholder data access, streamlining data-driven decision-making.

Thinknum Alternative Data | Jr Data Engineer New York | Jan-Mar 2022

- Developed ETL pipelines for web data extraction, eg info on >5mil Stackexchange users. Authored ORM parsers for Schema.org metadata.
- Resigned due to ethical concerns over military data scraping during Ukraine conflict.

University of Waterloo | CS Teaching Assistant; CS 135, CS 245, CS 330 (3x). 1/sem. Waterloo | Sep 2019 - Dec 2021

- Instructed business-relevant data engineering and analytics methodologies in CS 330, with focus on data modelling and orchestration.

Royal Bank of Canada, Toronto Innovation Lab | Software Engineer Summer Hire Toronto | May-Aug, 2017-2019

- Patented NLP and computer vision system for financial document analysis: `https://patents.google.com/patent/US11403465B2/en`.
- Developed natural-language interface for RBC market research, facilitating cross-vertical collaborations.
- Designed proof-of-concept conversational interfaces for onboarding and benefits selection, ultimately refined and adopted company-wide.

University of Pennsylvania | Head Teaching Assistant; LGIC 010 Philadelphia | Jan-May, 2017-2019

- Coauthored course **textbook** (free at `https://cis.upenn.edu/~weinstei/PQT.pdf`). Led TA team and curriculum development.

University of Pennsylvania | Senior Information Technology Advisor Philadelphia | Sep-May, 2015-2017

- Solved network and security problems for university members. Managed and trained other ITAs, led onboarding, and directed escalation.

Education

University of Waterloo | MMath | Computer Science Waterloo | Sep 2019 - Dec 2020

- **Selected graduate coursework, $\geq 90\%$ (equiv. A+) grade: Fairness, Accountability, and Transparency in Machine Learning**; **Machine Learning in Proteomics**; Automatic Sequences; Computer-Aided Verification

University of Pennsylvania | BA, Magna Cum Laude | Triple Major Philadelphia | Sep 2015 - May 2019

Logic, Information, and Computation; Computer Science; Philosophy and Science

- **Selected graduate coursework, $\geq$ A grade:** Logic and Computability; Networked Systems; Topology; Finite Model Theory
- Was selected as LGIC (Logic, Information, and Computation) departmental Convocation flagbearer in recognition of academic standing.

Expertise

- **Languages, Tools, and Clouds**: Julia, Python, SQL, NodeJS, C, K, OCaml, Java, Kotlin, Rust, • AWS, Azure, GCP • Docker, Kubernetes, Helm, Terraform, Kafka, Grafana
- **NLP**: LLMs, GPT, BERT, NLTK, spaCy, Conversational Design
- **AI/Machine Learning**: Pandas, NumPy, PyTorch, scikit-learn, Flux (Julia), Kubeflow, DVC, MLFlow, Spark, OpenCV, YOLO
- **Ethical AI**: ML fairness and dynamics (see `https://owainwest.com/fairness_dynamics.pdf`), LLM safety, Fairlearn, Fairness360
- **Data Engineering and Analytics**: Kafka, Airflow, Dagster, dbt, Snowflake, Spark, Singer.io, Airbyte, ADF, GraphQL, PostgreSQL, Redis, Streamlit, Looker, Superset